

DSA5208 Assignment 2

Singapore Rainfall Analysis & Prediction

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GitHub Repo: https://github.com/KohSiXing/DSA5208-Rainfall-Analysis

Rainfall Dataset: https://data.gov.sg/collections/2279/view

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Generative AI tools were used to support the preparation of this report and to refactor the original code. ChatGPT Terra 5.6 was used to improve the language, organisation, and clarity of the report. Claude Opus 5.5 was used to refactor the code after the team had developed the original implementation based on a defined data-analysis approach; this included improving code reusability and the presentation of outputs. The team determined the data-cleaning logic, exploratory data-analysis approach, machine-learning implementation, and interpretation of the results.

Abstract

This project uses Apache Spark to explore rainfall patterns across Singapore and estimate the probability that a station's rainfall will exceed a specified threshold over the next 30 minutes. The analysis draws on **48.5 million five-minute observations from 2017–2024**, covering **91 distinct station IDs**. Following data cleaning, seasonal, annual, spatial, and daily patterns were examined alongside short-term rainfall persistence. These findings guided the selection of **12 prediction features**.

A Gradient-Boosted Tree classifier was developed using Spark MLlib, with **2017–2022 used for training, 2023 for validation, and 2024 for testing**. Sampling reduced computational cost, and 27 model configurations were compared. The selected configuration was retrained on the combined training and validation samples before final testing. It achieved a **PR-AUC of 0.5706** and **ROC-AUC of 0.9070**, calculated across 5 rainfall thresholds pooled together. Current rainfall, the rainfall threshold, and recent 30-minute accumulation contributed most to the model. The results show the potential of recent rainfall history for short-term prediction, although probability calibration and performance at individual thresholds require further evaluation.

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1. Introduction

1.1 Background and objectives

This project analyses five-minute rainfall observations across Singapore from 2017–2024 using Apache Spark. It aims to understand rainfall patterns and estimate short-term rainfall risk through two objectives:

1. **Explore rainfall patterns:** Examine seasonal, annual, geographical, and time-of-day variation, alongside rainfall frequency, intensity, and persistence.
2. **Predict rainfall exceedance:** Develop and evaluate a Spark MLlib model that estimates the probability of rainfall exceeding a specified threshold in the next 30 minutes, given a station, prediction time, and threshold.

1.2 Project Workflow

The project consists of six main stages: data ingestion, data cleaning and storage, weather-pattern analysis, feature and label construction, chronological data splitting, and model development and evaluation. These stages are illustrated in Fig. 1.

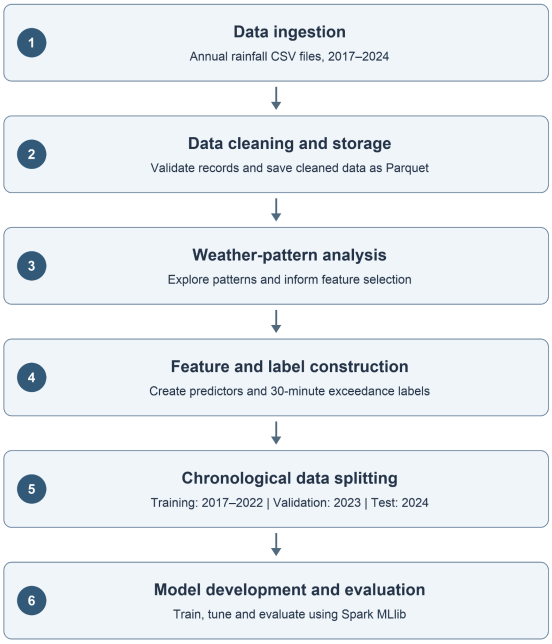


Fig 1: Project Workflow

2. Dataset and Computing Environment

2.1 Dataset Overview

This project uses the publicly available *Historical Rainfall across Singapore* datasets from data.gov.sg for 2017–2024. The datasets contain five-minute rainfall measurements, recorded in millimetres, from monitoring stations across Singapore.

Table 1 describes the 12 original columns. All columns were initially loaded as strings; the types shown reflect the conversions applied during data preparation.

Column	Data type	Description
`date`	Date	Date associated with the observation.
`timestamp`	Timestamp	Observation time.
`update_timestamp`	Timestamp	Update time recorded for the dataset entry.
`station_id`	String	Identifier of the monitoring station.
`station_name`	String	Name or location description of the station.
`station_device_id`	String	Identifier of the station's monitoring device.
`location_longitude`	Double	Station longitude in decimal degrees.
`location_latitude`	Double	Station latitude in decimal degrees.
`reading_update_timestamp`	Timestamp	Update time recorded for the rainfall reading.
`reading_value`	Double	Rainfall accumulated over the five-minute interval, in millimetres.
`reading_type`	String	Description of the rainfall measurement type.
`reading_unit`	String	Measurement unit, recorded as millimetres (`mm`).

Table 1: Dataset columns, data types, and descriptions

The combined dataset contains **48,538,713 rows and 12 original columns**, covering **91 distinct station IDs** across the study period. The number of reporting stations varies by year; not all 91 stations contributed observations throughout 2017–2024.

2.2 Computing Environment

Data processing and model development were performed on Google Cloud Dataproc using PySpark with Apache Spark 3.5.3 and YARN for resource management. The cluster comprised one master node for coordination and three worker nodes for distributed processing. Each node was a separate virtual machine, with the hardware specifications shown in Table 2.

Node role	**Number of nodes**	**Machine type**	**vCPUs per node**	**RAM per node**	**Boot disk per node**
Master	1	`n2-standard-4`	4	16 GiB	100 GB standard persistent disk
Worker	3	`n2-highmem-4`	4	32 GiB	100 GB balanced persistent disk
Total	**4**	—	**16 vCPUs overall**	**112 GiB overall**	**400 GB overall**

Table 2: Cluster hardware configuration

The three worker nodes together provided **12 vCPUs and 96 GiB of RAM** for distributed processing, before accounting for system overhead. RAM is the working memory used during computation, while boot disks provide storage for the operating system and local files.

Component	Configuration
Cluster name	`rainfall-dev`
Region / zone	`asia-southeast1` / `asia-southeast1-b`
Dataproc image	`2.2-debian12`
Processing framework	Apache Spark 3.5.3 through PySpark
Resource manager	YARN
Notebook interface	Jupyter, with Component Gateway enabled
Spark time zone	`Asia/Singapore`
Maximum idle duration	30 minutes

Table 3: Platform and software configuration

Google Cloud Storage held the raw CSV files, cleaned Parquet data, prepared modeling datasets, and model outputs. This storage was separate from the nodes' boot disks. Cleaned data was saved in Parquet format for subsequent analysis and modeling.

Spark divided the data into partitions that could be processed in parallel across the worker nodes. For feature engineering and modeling, 128 shuffle partitions were configured, and adaptive query execution allowed Spark to adjust processing during execution. The sampled modeling datasets were divided into 32 partitions; these partitions were processed as computing resources became available.

To reduce repeated computation, the full prepared feature dataset was temporarily stored on worker disks. Smaller modeling datasets were cached in RAM where possible, with disk storage used when necessary.

3. Data Inspection and Preparation

Before analysis and modeling, the raw datasets were inspected and cleaned to ensure consistent data types, station information, and rainfall measurements. This involved checking data quality, correcting station names and coordinates, and saving the prepared data for subsequent use.

3.1 Schema and Data-Quality Checks

All 12 original columns were initially loaded as strings. Dates and timestamps were converted to date and timestamp types, while rainfall values and geographic coordinates were converted to double-precision numeric values. Timestamps were interpreted in Singapore time.

Data-quality checks covered missing values, duplicates, measurement consistency, and station information. Table 4 summarises the checks, findings, and actions taken.

Category	**Checks performed**	**Findings and actions**
Missing values	Checked for nulls, blank strings, and common missing-value markers such as `NA`, `NULL`, and `NONE`.	None were found; no imputation or replacement was required.
Duplicates	Checked for exact duplicate rows and repeated station–timestamp combinations.	None were found; no deduplication was required.
Measurement consistency	Checked rainfall measurement types and units across records.	All records used the same five-minute rainfall measurement type and millimetre unit; no conversion was required.
Station information	Checked station names and geographic coordinates for inconsistencies.	Naming variations and two coordinate pairs for S113 were identified. Names

		and coordinates were standardised as described in Section 3.2.

Table 4: Data-quality checks and preparation decisions

3.2 Cleaning Station Names and Coordinates

The checks in Section 3.1 found that 26 station IDs had multiple names. These included station IDs appearing as names, inconsistent capitalisation, and abbreviations. Table 5 illustrates these issues and the names adopted after cleaning.

Station ID	**Issue identified**	**Original names**	**Standardised name**
S06	Station ID used as a name	Paya Lebar; S06	Paya Lebar
S215	Inconsistent capitalisation	GEYLANG EAST CENTRAL; Geylang East Central	Geylang East Central
S216	Abbreviation and station ID used as names	Ang Mo Kio Ave 10; Ang Mo Kio Avenue 10; S216	Ang Mo Kio Avenue 10

Table 5: Examples of station-name inconsistencies and their corrections

The following steps were taken:

1. **Standardise station names:** Correct capitalisation, expand “Ave” to “Avenue”, and replace IDs used as names with available descriptive names.
2. **Preserve the original names:** Store cleaned names in `station_name_clean`, retaining the original column. Use station IDs to group observations consistently.
3. **Standardise S113’s coordinates:** S113 also had two coordinate pairs approximately 39 metres apart. The later pair, which covered most observations, was adopted as its standard location.

3.3 Rainfall Distribution and Extreme Values

Five-minute rainfall readings ranged from **0 to 37.2 mm**, with a mean of **0.0253 mm**. The approximate 95th percentile was zero, indicating that most recorded intervals were dry.

Among positive readings, small rainfall amounts were common, while large amounts were rare. Fig. 2 shows this strongly right-skewed distribution using a logarithmic frequency axis to make the less frequent, larger readings visible.

The maximum reading of **37.2 mm** was retained, as an unusually large value alone does not establish a measurement error. Further evidence would be needed to justify its removal.

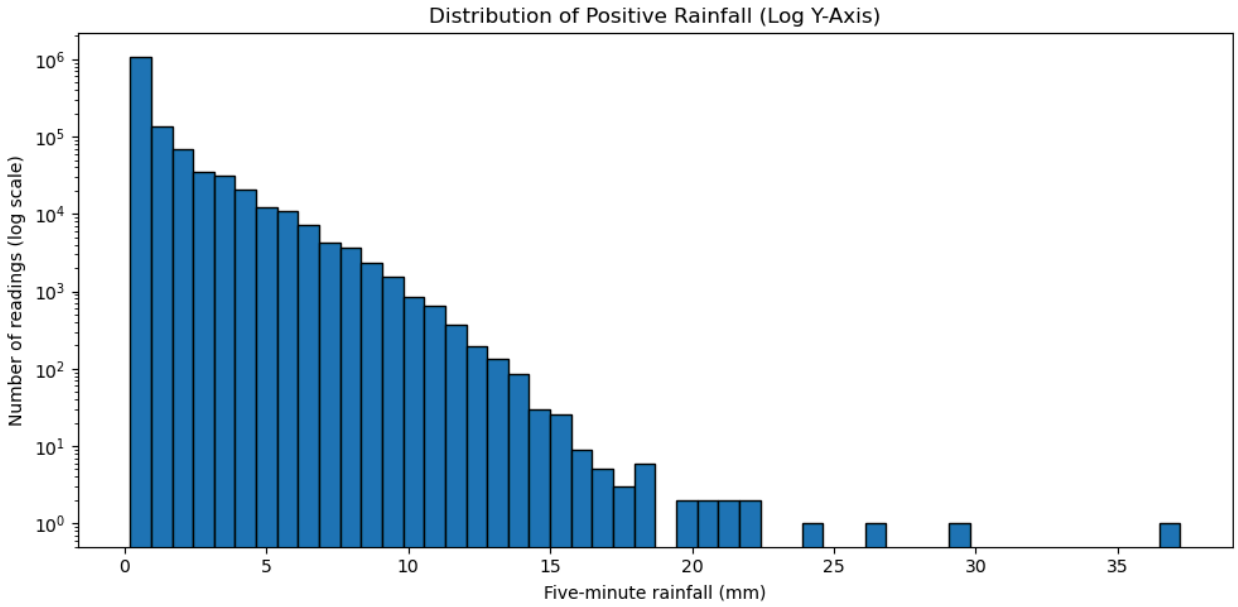


Fig 2: Positive five-minute rainfall distribution (log scale), 2017–2024.

3.4 Prepared Data Output

The cleaned dataset was saved in Parquet format in Google Cloud Storage for use in weather-pattern analysis and feature engineering. This preserved the converted data types and standardised station information, allowing both tasks to use the same prepared data without repeating the cleaning steps.

4. Task 1: Rainfall Patterns Within and Across Years

4.1 Overview of the Analysis

To understand how rainfall varies across Singapore, we first examine the typical monthly pattern and how it differs between years. We then compare regions and times of day to explore where and when rainfall occurs. Wetter periods are examined further to distinguish between more frequent rain and heavier rainfall. Finally, we investigate whether recent readings provide useful information for predicting rainfall.

Analysis	**Main question**
Seasonal patterns	Which months are typically wetter or drier?
Annual variation	How do rainfall amounts and monthly patterns differ across 2017–2024?
Regional differences	Do wetter and drier periods occur at similar times across regions?
Time-of-day patterns	How does rainfall vary throughout the day and across months?
Frequency and intensity	Does a wetter month experience rain more often, or in larger amounts?
Rainfall persistence	How strongly is current rainfall related to recent readings?

Table 6: Questions addressed in the rainfall analysis

4.2 Seasonal Rainfall Patterns

To identify wetter and drier months, monthly rainfall was estimated by multiplying the mean observed five-minute rainfall by 288 intervals per day and the number of days in the month. Each calendar month's estimates were then averaged across 2017–2024.

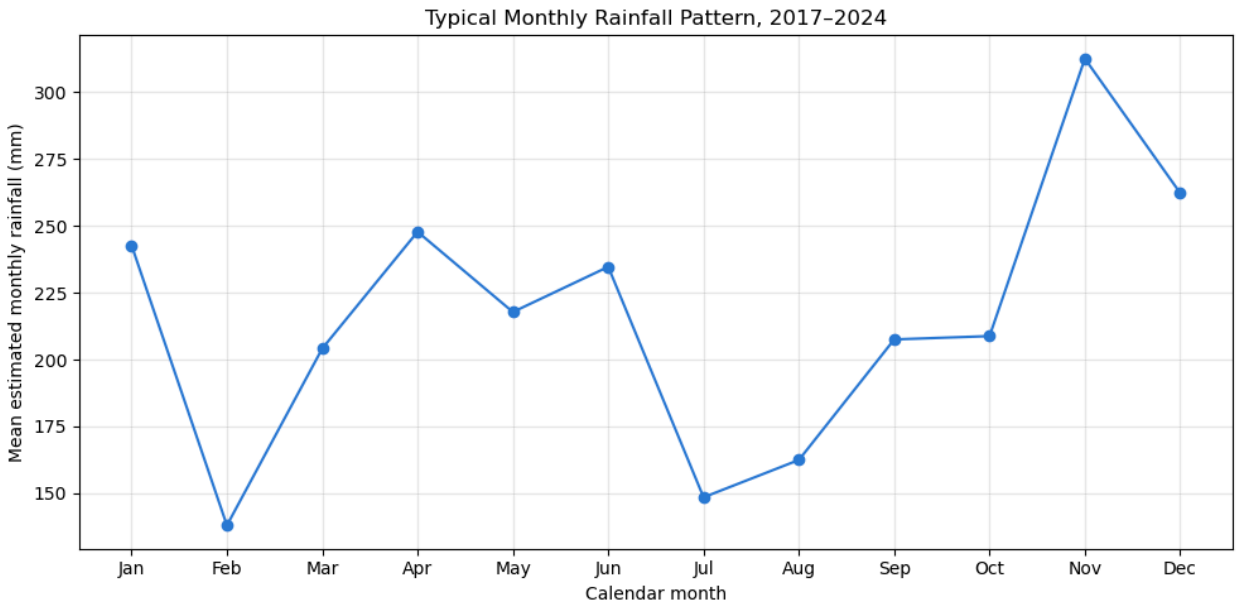


Fig 3: Average estimated monthly rainfall, 2017–2024

November had the highest estimated monthly rainfall at **312.5 mm**, followed by December at **262.3 mm**. February had the lowest estimate at **138.0 mm**, while July and August were also relatively dry at **148.5 mm** and **162.5 mm**, respectively.

Rainfall generally increased from February to April, declined in July, and rose again towards the November peak. This suggests a seasonal pattern, with wetter conditions towards the end of the year. However, the profile represents an eight-year average and does not imply that November was the wettest month in every year.

4.3 Differences Across 2017–2024

Section 4.2 showed the typical seasonal pattern, but averaging across years can conceal differences between individual years. We therefore compared annual rainfall estimates and examined the monthly timeline to determine how consistently this pattern appeared throughout 2017–2024.

Annual rainfall was estimated by multiplying each year's mean observed five-minute rainfall by the expected number of intervals in that year, accounting for leap years.

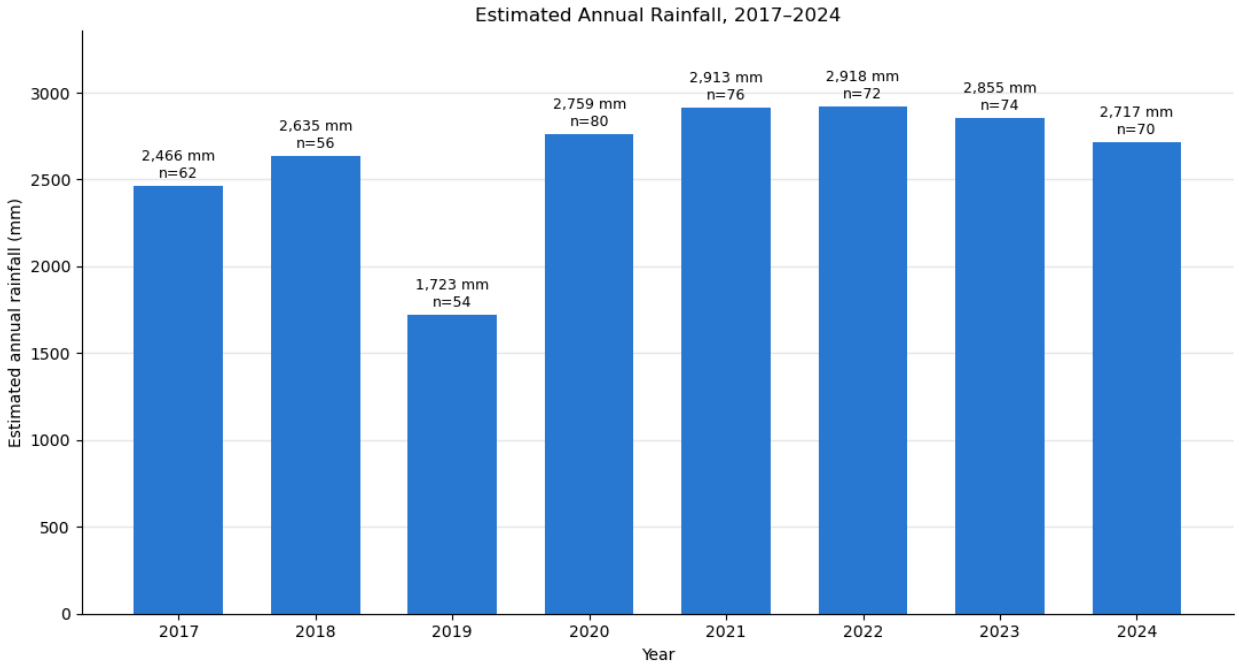


Fig 4: Estimated annual rainfall and reporting station counts, 2017–2024

The lowest annual estimate occurred in **2019, at approximately 1,723 mm**. Rainfall increased sharply in 2020 and remained relatively high through 2023. The highest estimate occurred in **2022, at approximately 2,918 mm**, closely followed by **2021, at 2,913 mm**. In 2024, the estimate declined to approximately **2,717 mm**.

To examine the monthly variation behind these annual estimates, Fig 5 presents all 96 months in chronological order.

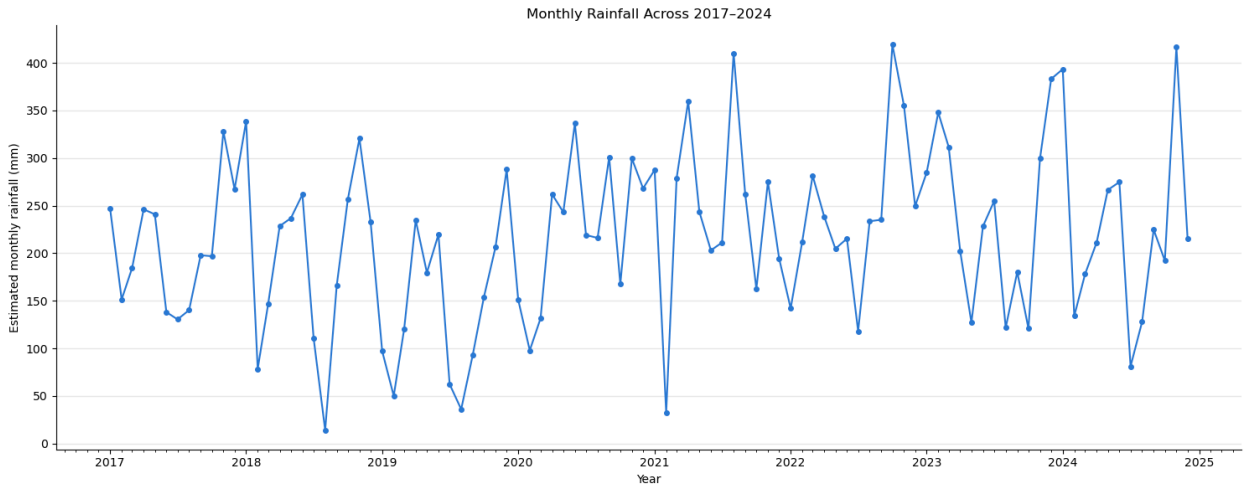


Fig 5: Estimated monthly rainfall across 2017–2024

Several low-rainfall months occurred in 2019, consistent with its lower annual estimate. Pronounced monthly peaks also appeared in other years, including estimates above 400 mm in 2021, 2022, and 2024. Although November was the wettest month on average, individual years differed in the timing and magnitude of their peaks.

These estimates assume that available readings represent the full period and give more weight to stations with more observations. The number of reporting stations ranged from **54 to 80** across years, so changes in coverage may influence the comparisons. The results describe variation during the study period rather than establish a long-term climate trend.

4.4 Spatial Variation

The annual and monthly comparisons showed when rainfall increased or decreased across the dataset. We next examined whether these changes were shared across Singapore or differed by location.

Stations were assigned to Central, North, South, East, and West groups using their coordinates relative to a selected central reference point. These were custom geographic groupings rather than official regional boundaries. Rainfall was summed for each station within each month, then averaged across reporting stations in each region.

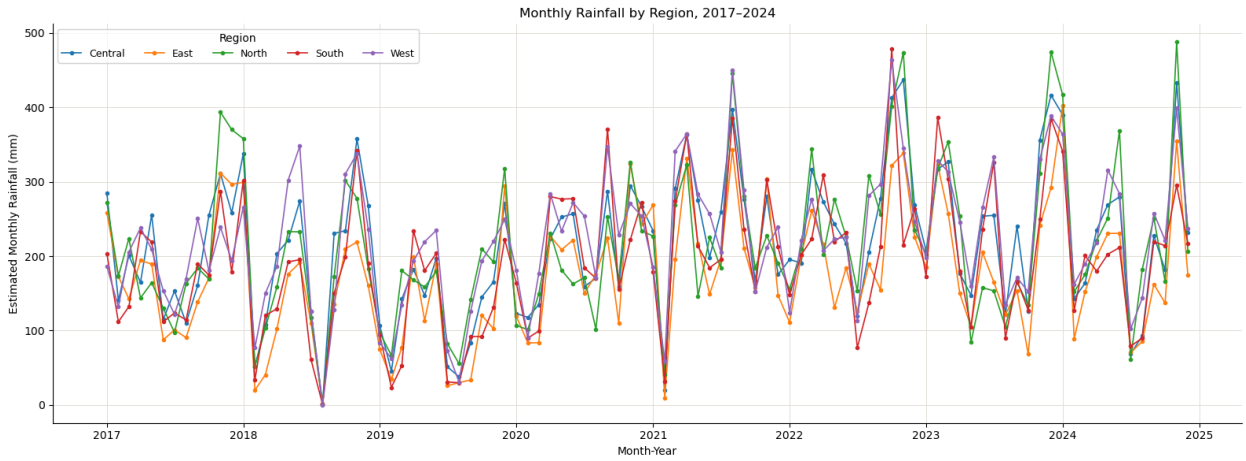


Fig 6: Average observed monthly rainfall by region, 2017–2024

The regions often experienced peaks and troughs at similar times, suggesting that many wetter and drier periods were shared across Singapore. Several months in 2019 had low rainfall across all five regions, supporting the annual findings in Section 4.3.

However, the magnitude of rainfall differed between regions. The North showed several prominent peaks towards the end of 2022–2024, while the East often recorded lower monthly amounts. No region consistently had the highest rainfall throughout the study period.

Regional rainfall was calculated by averaging the observed monthly totals across reporting stations, giving each station equal weight. These totals were not adjusted for missing readings, so incomplete records may underestimate rainfall. Near-zero values in parts of 2018 should therefore be checked for missing observations before being interpreted as exceptionally dry conditions.

4.5 Time of Day, Frequency, and Intensity

4.5.1 Time-of-Day Patterns

Following the regional comparison, we examined when rainfall occurred during the day. Mean five-minute rainfall was calculated for each month–hour combination across stations and years, using Singapore time.

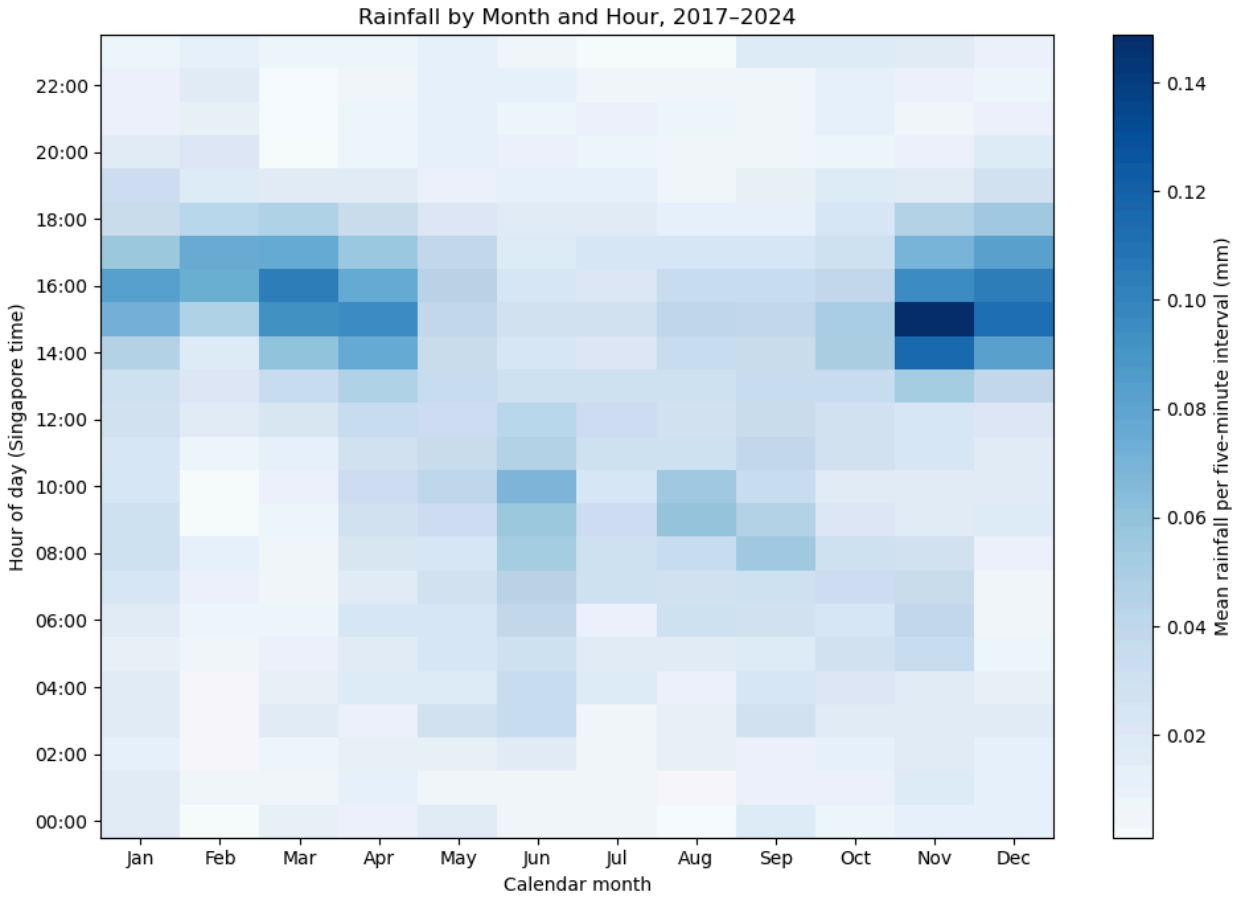


Fig 7: Mean five-minute rainfall by month and hour, 2017–2024

Afternoon rainfall was prominent around **14:00–17:00**, particularly in January–April and November–December, with the strongest peak around **15:00 in November**. Morning rainfall was relatively more prominent in June, August, and September. These patterns support considering month and hour as prediction features.

4.5.2 Rainfall Frequency and Intensity

Higher average rainfall may reflect more frequent rain, heavier rain, or both. We therefore compared the percentage of wet intervals with the mean rainfall amount during those intervals.

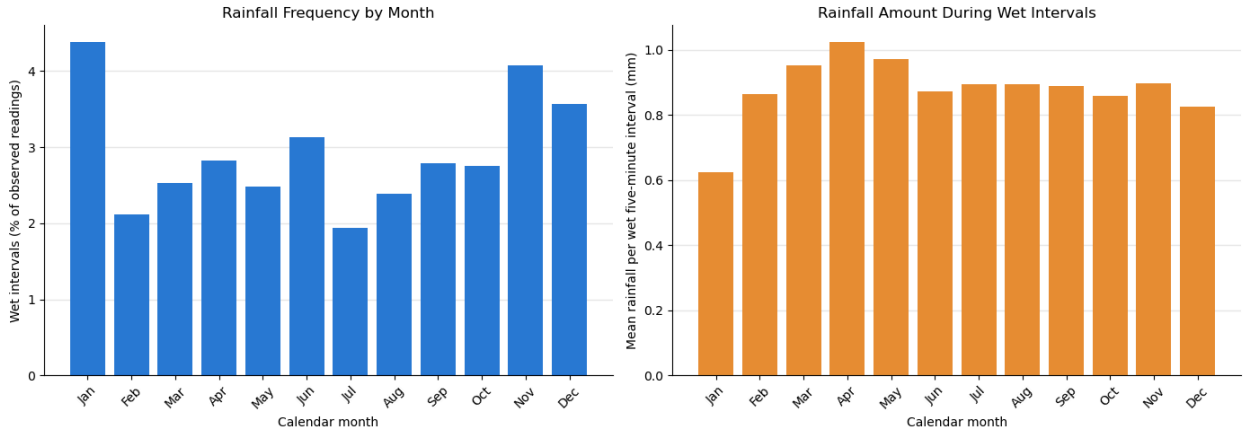


Fig 8: Monthly rainfall frequency and mean amount per wet interval.

January had the highest wet-interval frequency (**4.38%**) but the lowest mean wet-interval amount (**0.62 mm**). April had the highest mean amount (**1.02 mm**), despite a lower frequency (**2.83%**). November combined relatively high frequency (**4.08%**) and intensity (**0.90 mm**), helping explain its high monthly rainfall estimate.

These results show that frequent rain does not necessarily mean heavier rain. The percentages describe observed five-minute intervals, not rainy days.

4.6 Rainfall Persistence and Implications for Prediction

After examining when rainfall occurs and how its frequency and intensity vary, we investigated whether current rainfall is related to recent readings at the same station. Correlations were calculated at different time lags, using observations with the expected time separation.

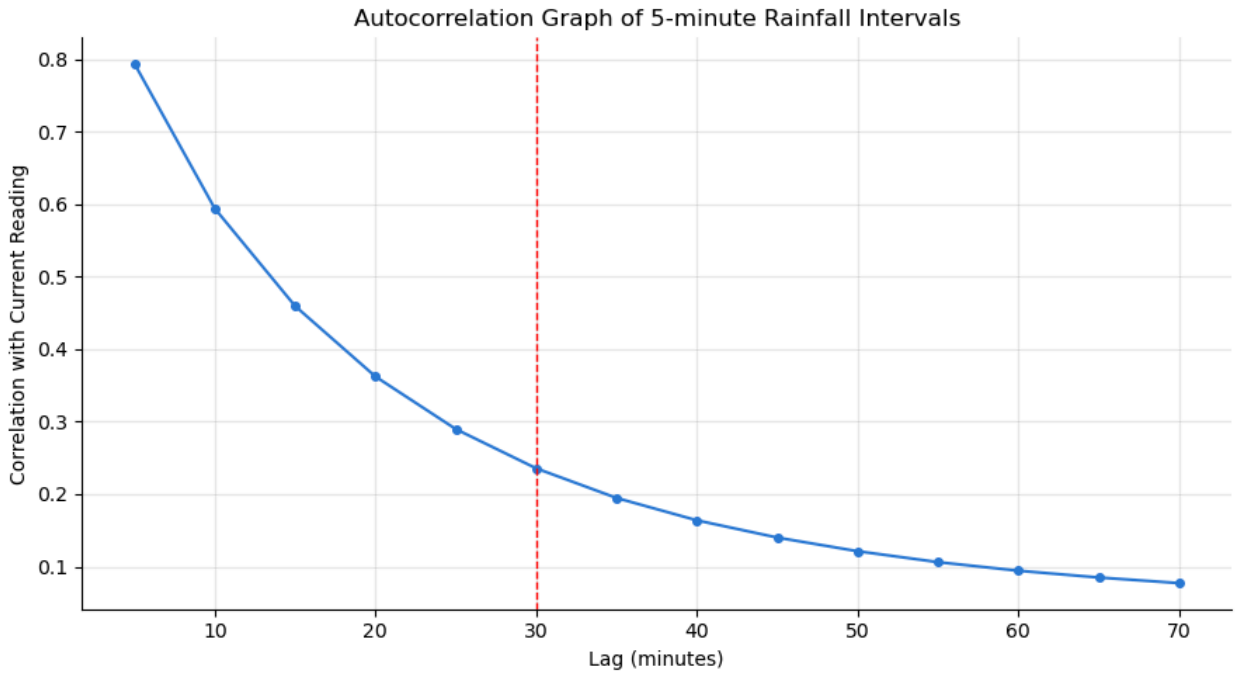


Fig 9: Correlation between rainfall readings at different time lags

The correlation was approximately **0.79 at five minutes**, falling to **0.24 at 30 minutes** and **0.09 at one hour**. This indicates strong short-term persistence that weakens as readings become further apart.

These findings support using recent rainfall readings and accumulated rainfall as candidate features for Task 2. However, the correlations compare individual five-minute readings; they do not directly measure how accurately the next 30-minute rainfall total can be predicted.

5. Task 2: Prediction Methodology

5.1 Prediction Target and Inputs

The model estimates the probability that a station's accumulated rainfall over the next 30 minutes exceeds a specified threshold.

Target probability = P(next 30-minute rainfall total > threshold)

The future total, `future_30m`, is the sum of the next six five-minute readings. It is used to create the label, not as a model input. Table 7 summarises the prediction inputs and target labels.

Item	**Definition**
Station	Identifies the location and rainfall history used for prediction.
Prediction time	Determines the calendar features and available rainfall readings.
Rainfall threshold	Amount to exceed: **0, 1, 2.5, 5, or 10 mm** in the implemented experiments.
Label 1	Future 30-minute rainfall is **greater than** the threshold.
Label 0	Future 30-minute rainfall is **equal to or below** the threshold.

Table 7: Prediction inputs and target labels

Each sampled station–time observation was expanded into five examples, one per threshold, allowing the model to estimate exceedance probabilities for different rainfall amounts.

5.2 Feature Engineering

The findings in Task 1 guided the selection of 12 model inputs. Location and calendar features capture spatial and temporal variation, while current and historical rainfall features describe recent conditions. Table 8 summarises these features, where **t** represents the prediction time.

Feature group	**Column**	**Spark data type**	**Definition**
Location	`location_latitude`, `location_longitude`	Double	Station coordinates.
Calendar	`hour`, `month`	Integer	Hour and month extracted from the observation timestamp in Singapore time.
Current rainfall	`reading_value`	Double	Five-minute rainfall reading at t.
Lagged rainfall	`lag_5m`, `lag_30m`, `lag_1h`	Double	Readings recorded 5, 30, and 60 minutes before t, respectively.
Recent accumulation	`rain_cumsum_30m`	Double	Sum of six readings from t − 25 minutes through t.
Recent accumulation	`rain_cumsum_1h`	Double	Sum of twelve readings from t − 55 minutes through t.
Recent frequency	`wet_count_30m`	Long	Number of positive readings among the six readings from t − 25 minutes through t.

Rainfall threshold	`thr`	Double	Threshold used to estimate the exceedance probability.

Table 8: Model features, data types, and definitions

Historical features were calculated separately for each station, with timestamp checks to ensure the intended time gaps. Both accumulation windows include the current reading. Station ID was used to construct each station's history but was not included directly as a model feature.

All rainfall features used observations at or before the prediction time. Future readings were reserved for constructing the target label.

5.3 Data Quality and Leakage Prevention

5.3.1 Completeness of Features and Target

Before modeling, observations were checked to ensure that historical features and future rainfall totals were calculated from complete records. Table 9 summarises these checks.

Check	**Relevant columns**	**Requirement**
Complete rainfall history	Current rainfall, lag features, rainfall sums, and wet-interval count	13 readings from t − 60 minutes through t, each 5 minutes apart.
Correct lag times	`lag_5m`, `lag_30m`, `lag_1h`	Earlier readings must be exactly 5, 30, and 60 minutes before t, respectively.
Complete target window	`future_30m`, used to create `label`	6 readings at t + 5, t + 10, …, t + 30 minutes.
No missing values	All features except `thr`, plus `future_30m`	Exclude rows with null values in these columns. The threshold (`thr`) is added later.

Table 9: Data-quality checks for features and target construction

For a prediction at **10:00**, the historical check requires the **09:00 reading for `lag_1h`** and **12 readings from 09:05 through 10:00 for the 1-hour sum**. This combined history also supports the shorter historical features.

After these checks, **47,000,882 station–time observations** remained eligible before split-boundary filtering and sampling.

5.3.2 Preventing Future Information from Entering Model Inputs

Rainfall inputs included the current reading and earlier readings from the same station. The future total, `future_30m`, was used to create the target label and excluded from the model inputs. This ensured that the model learned to predict the outcome without being given the future rainfall used to determine it.

5.4 Data Splitting and Sampling

The eligible observations were split chronologically so that models were trained on earlier years, tuned using 2023 data, and evaluated on 2024 data. Sampling was applied to reduce computation time. Table 10 summarises the datasets.

Dataset	**Period**	**Sampling fraction**	**Purpose**
Training	2017–2022	0.5%	Fit the model.
Validation	2023	5%	Compare configurations and select hyperparameters.
Test	2024	10%	Evaluate the selected model on later observations.

Table 10: Chronological splits and sampling fractions

Sampling reduced the computational cost of repeated training and evaluation. Approximately 47 million station–time observations passed the completeness checks. Expanding all observations into 5 threshold-specific examples would produce approximately 235 million rows before split-boundary exclusions. Sampling made the experiments more manageable, although fewer rare heavy-rainfall examples were retained.

The data was prepared for modeling in the following steps:

1. **Keep target windows within each split:** The 30-minute target window had to end before the next period began. For example, a prediction at 23:45 on 31 December 2022 was excluded from training because its target extended into 2023.
2. **Sample eligible observations:** Features and future rainfall totals were calculated before sampling to preserve complete time windows. Observations were then sampled without replacement using seed **42**, independently of their target labels.
3. **Create threshold-specific examples:** Each sampled station–time observation was expanded into **5 examples**, using thresholds of **0, 1, 2.5, 5, and 10 mm**.
4. **Check class representation:** Positive and negative labels were counted for each threshold in each dataset to confirm that both classes were present.

The samples contained **169,649 training**, **333,593 validation**, and **633,464 test** station–time observations before threshold expansion. Each observation contributed 1 example per threshold. Table 11 presents the positive-label proportions.

Rainfall threshold	Training positive (%)	Validation positive (%)	Test positive (%)
> 0 mm	6.06	6.69	6.36
> 1 mm	2.41	2.66	2.57
> 2.5 mm	1.42	1.54	1.47
> 5 mm	0.86	0.90	0.84
> 10 mm	0.41	0.42	0.38

Table 11: Positive-label proportions by rainfall threshold

Positive outcomes became rarer as the rainfall threshold increased. Only **0.41%** of training examples exceeded 10 mm, highlighting the strong class imbalance. This informed the choice of evaluation metrics, with PR-AUC used for model selection and ROC-AUC reported as a supplementary measure, as discussed in Section 5.5.

5.5 Model Development, Evaluation Metrics, and Configuration Selection

A Gradient-Boosted Tree (GBT) classifier was developed using Spark MLlib. GBT was chosen because it can capture nonlinear relationships and interactions between features. For example, the relationship between recent rainfall and future exceedance may depend on the rainfall threshold or season. Tree-based models also do not require feature scaling, allowing rainfall amounts, coordinates, and calendar variables to be used together.

Spark MLlib supports distributed GBT training within the project's computing environment. Feature-importance scores also help explain which inputs contributed most to the fitted model.

The pipeline consisted of 2 stages: `VectorAssembler` combined the 12 inputs into a feature vector, and `GBTClassifier` estimated the probability that the next 30-minute rainfall total would exceed the specified threshold.

5.5.1 Hyperparameter Tuning

Models were trained on the sampled 2017–2022 data and evaluated on the 2023 validation sample. A grid search tested **27 configurations**, formed from the combinations of tree depth, learning rate, and boosting iterations shown in Table 12.

Parameter	**Values tested**	**Purpose**
Maximum depth (`maxDepth`)	3, 5, 7	Controls the complexity of each tree.
Learning rate (`stepSize`)	0.01, 0.05, 0.1	Controls the contribution of each additional tree.
Boosting iterations (`maxIter`)	50, 100, 150	Sets the number of boosting iterations.
Random seed (`seed`)	42	Provides a fixed seed for reproducibility.

Table 12: GBT hyperparameter search

The same training and validation samples were used for all configurations. Model-fitting time was also recorded to compare computational cost.

5.5.2 Evaluation Metrics

The class imbalance identified in Section 5.4 made accuracy alone unsuitable for assessing performance. PR-AUC was therefore used as the main selection metric, with ROC-AUC reported as a supplementary measure. Table 13 explains their roles.

Metric	**Interpretation**	**Role**
Precision–recall AUC (PR-AUC)	Summarises precision and recall across probability cutoffs, assessing the trade-off between catching exceedances and avoiding false alarms.	Main metric for comparing configurations.
Receiver operating characteristic AUC (ROC-AUC)	Summarises the true positive rate against the false positive rate across probability cutoffs.	An additional measure of how well the model separates positive from negative cases

Table 13: Model evaluation metrics

Both metrics assess prediction rankings across probability cutoffs. They do not directly assess whether predicted probabilities match observed event frequencies.

5.5.3 Selecting the Best Model Configuration

Each of the 27 configurations was assessed using the same procedure:

1. **Predict on validation data:** Apply the trained model to the 2023 validation sample.
2. **Evaluate each rainfall threshold:** Calculate PR-AUC separately for thresholds of **0, 1, 2.5, 5, and 10 mm**.
3. **Calculate the average:** Add the 5 PR-AUC scores and divide by 5, giving each threshold equal weight.
4. **Compare configurations:** Select the configuration with the highest average validation PR-AUC.

This procedure selected **1 GBT model that accepts different rainfall thresholds**, rather than a separate model for each threshold. ROC-AUC was also recorded but did not determine the

selected configuration. The 2024 test sample was reserved for final evaluation and was not used to choose the model configuration.

6. Task 2: Prediction Results and Discussion

6.1 Validation Results and Selected Configuration

The 27 GBT configurations were compared using their mean validation PR-AUC across the 5 rainfall thresholds. Table 14 presents the 3 highest-scoring configurations.

Maximum depth	**Learning rate**	**Iterations**	**Mean PR-AUC**	**Mean ROC-AUC**	**Model training time (seconds)**
3	**0.01**	**150**	**0.4763**	**0.8492**	**68.5**
3	0.05	50	0.4761	0.8563	26.3
5	0.01	150	0.4757	0.8921	104.8

Table 14: Leading GBT configurations on the validation sample

The selected configuration used a maximum depth of **3**, learning rate of **0.01**, and **150 iterations**. It achieved the highest mean validation PR-AUC of **0.4763**.

However, the second configuration achieved a very similar PR-AUC with a shorter fitting time of **26.3 seconds**. This suggests that a less computationally expensive configuration may offer comparable performance, although repeated experiments would be needed to assess the reliability of the small difference.

The selected configuration did not have the highest ROC-AUC among these alternatives. This reflects the use of PR-AUC as the selection criterion, prioritising performance on the relatively rare positive outcomes.

6.2 Final Model Training and Test Performance

The selected GBT configuration was retrained on the combined training and validation samples from **2017–2023**, then evaluated on the held-out **2024 test sample**. Table 15 summarises the results.

Measure	**Result**
Test PR-AUC	0.5706
Test ROC-AUC	0.9070
Positive-label proportion across pooled test examples	2.32%
Model training time	228.3 seconds

Table 15: Final model performance on the 2024 test sample

The model achieved a **PR-AUC of 0.5706**, substantially above the random-ranking baseline of **0.0232**, based on the pooled test data's **2.32% positive rate**. This indicates a better balance between detecting rainfall above the threshold and avoiding false alarms across probability cutoffs.

The **ROC-AUC of 0.9070** also indicates that the model generally assigns higher scores to positive cases than negative cases. Neither score represents classification accuracy. Precision and recall at a chosen cutoff are needed to assess how many alerts are correct and how many actual events are detected.

6.3 Final-Model Feature Importance

Feature-importance scores were examined to identify the main contributors to the final model's predictions following retraining on the combined training and validation samples. Fig 10 presents these scores.

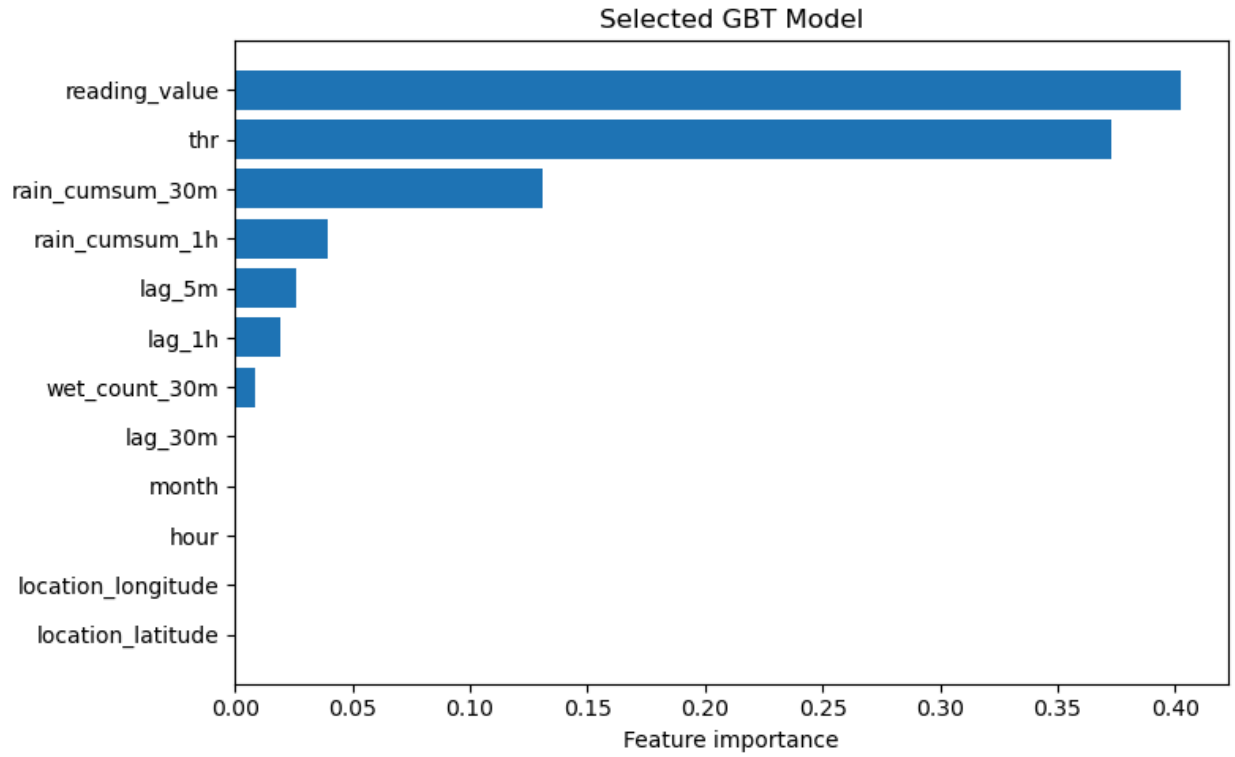


Fig 10: Feature importance of the final GBT model

Current rainfall recorded the highest importance (**40.2%**), followed by the rainfall threshold (**37.3%**) and recent 30-minute rainfall accumulation (**13.1%**). Together, these features accounted for approximately **90.6%** of the total importance.

The prominence of current and recent rainfall is consistent with the short-term persistence identified in Section 4.6. The threshold also contributed substantially, as it defines the rainfall amount to be exceeded.

Location, hour, month, and the 30-minute lag received zero importance in the final model. Although motivated by the exploratory analysis, these features provided no measured contribution to the fitted trees. This does not imply that they lack predictive value generally, as related features may capture similar information. Feature-importance scores describe the model's use of its inputs, rather than causal effects or contributions to accuracy.

6.4 Example Predictions

The prediction function accepts a station ID, prediction time, and rainfall threshold. It selects the latest available prepared reading at or before the requested time and estimates the probability that rainfall will exceed the threshold over the following 30 minutes.

Table 16 illustrates the function's outputs alongside the observed outcomes. These examples demonstrate how predictions are interpreted; overall performance is assessed using the test metrics in Section 6.2. All times are in Singapore time.

Station	**Requested time**	**Latest reading at or before requested time**	**Target window ends**	**Threshold (mm)**	**Predicted probability**	**Recorded future total (mm)**	**Actual outcome**
S06	7 Nov 2024, 16:52	16:50	17:20	0.5	71.68%	0.400	Did not exceed threshold
S116	7 Jan 2024, 16:57	16:55	17:25	0.5	45.88%	0.400	Did not exceed threshold
S221	7 Nov 2024, 16:30	16:30	17:00	1.0	83.95%	1.188	Exceeded threshold

Table 16: Example predictions and observed outcomes

1. **Prediction window:** Each window begins immediately after the selected reading and ends 30 minutes later on the same date. For S06, the request at 16:52 uses the 16:50 reading, with the future total calculated from the 6 readings timestamped **16:55–17:20**.
2. **Predicted probabilities and outcomes:** S221 received a probability of **83.95%** and subsequently exceeded its threshold. S06 received a probability of **71.68%**, but its rainfall remained below the threshold. Probabilities express uncertainty, so a high predicted probability does not guarantee that an event will occur. These individual outcomes alone cannot establish the quality of the probability estimates.
3. **Threshold coverage:** The **0.5 mm** threshold used for S06 and S116 was not among the 5 thresholds used during training and evaluation. The function accepts this value, but predictive performance at this threshold remains unverified.

7. Discussion, Limitations, and Future Work

7.1 Data and Rainfall Analysis Limitations

The analysis describes rainfall patterns across 2017–2024. Table 17 highlights 2 considerations when interpreting these findings.

Limitation	**Effect on interpretation**	**Suggested improvement**
Observation and station coverage	Task 1 used available readings, while the number of reporting stations varied across years. Missing observations and changes in station coverage may influence rainfall comparisons.	Assess coverage by station and month, and repeat comparisons using consistently reporting stations.
Limited study period	The 8-year period supports comparisons within 2017–2024 but does not establish a long-term climate trend.	Extend the analysis to a longer period with comparable observation coverage.

Table 17: Main data and analysis limitations

7.2 Prediction Limitations and Improvements

Table 18 highlights 3 limitations most relevant to interpreting the model's performance.

Limitation	**Explanation**	**Suggested improvement**
Reliability of predicted probabilities	PR-AUC and ROC-AUC assess prediction rankings but do not confirm whether predicted probabilities match observed outcomes. For example, predictions near 70% should correspond to events occurring approximately 70% of the time.	Assess probability quality using a reliability diagram and Brier score.
Test results combined across thresholds	The reported scores combine all 5 rainfall thresholds and may conceal weaker performance at individual thresholds.	Report PR-AUC and ROC-AUC separately for each rainfall threshold.
Validation using only 2023	Model configurations were compared using 1 validation year, so the selected configuration may depend on that year's rainfall patterns.	Use several chronological training–validation splits, keeping 2024 separate for final testing.

Table 18: Main prediction limitations and suggested improvements

7.3 Further Extensions

Beyond the improvements discussed in Section 7.2, the project could be extended in 3 directions:

1. **Compare alternative models:** Evaluate GBT against a simple baseline, logistic regression, and random forests using the same chronological splits and metrics. This would establish whether GBT offers an advantage over simpler approaches.
2. **Include additional weather information:** Add recent rainfall from neighbouring stations and variables such as humidity, temperature, and wind, where available. These may provide information about approaching rainfall that is not captured by the station's own history.
3. **Check consistency across thresholds:** For the same station and prediction time, the probability of exceeding a higher rainfall threshold should not exceed the probability of exceeding a lower threshold. This property could be tested across a wider range of thresholds, as the current model does not explicitly enforce it.

8. Conclusion

This project used Apache Spark to analyse Singapore's five-minute rainfall observations from **2017–2024** and develop a model for predicting rainfall above a specified threshold over the next 30 minutes.

The rainfall analysis informed feature selection: seasonal and daily patterns motivated month and hour features, spatial variation motivated station coordinates, and short-term persistence motivated current readings, rainfall lags, recent accumulations, and wet-interval counts.

A Spark MLlib GBT model was selected through hyperparameter tuning and retrained on the combined training and validation samples. It achieved a pooled test **PR-AUC of 0.5706** and **ROC-AUC of 0.9070**, indicating useful discrimination between cases that exceeded the rainfall threshold and those that did not. Current rainfall, the requested threshold, and recent 30-minute accumulation contributed most to the final model.

These results demonstrate the potential of historical rainfall for short-term prediction. Evaluation by rainfall threshold, probability calibration, and validation across additional time periods would strengthen confidence in practical use.

References:

Apache Software Foundation. (2024). BinaryClassificationEvaluator — PySpark 4.2.0 documentation. https://downloads.apache.org/spark/docs/4.2.0/api/python/reference/api/pyspark.ml.evaluation.BinaryClassificationEvaluator.html

Apache Software Foundation. (2026). GBTClassifier — PySpark 4.2.0 documentation. https://spark.apache.org/docs/latest/api/python/reference/api/pyspark.ml.classification.GBTClassifier.html

Assistant Professor Kok, S. (2022). Course title: BT4013 Analytics for Capital Market Trading and Investment [Lecture notes]. Department of Information Systems and Analytics (DISA), School of Computing (SoC) , National University of Singapore.

National Environment Agency. (2026). *Historical Rainfall across Singapore* [Data set collection]. data.gov.sg. https://data.gov.sg/collections/2279/view